CLINICAL BIOINFORMATICS · AGENTIC CODING

David Chelimo

Clinical Bioinformatics Scientist II — Labcorp

Boston, MA (US)
~10 years in genomics
Labcorp — May 2026

Clinical bioinformatics scientist with nearly a decade of experience in cancer genomics and clinical NGS. Proficient in Python and SQL, comfortable with Bash, R, and Java. Deep experience in annotated genomic profiles, somatic variant review, and QA/QC in high-complexity regulated environments. Actively focused on agentic coding workflows, AI coding assistants, and moving toward a bioinformatics software engineering role.

EXPERIENCE

Clinical Bioinformatics Scientist II
May 2026 – Present

Labcorp · Remote (US)

- Support clinical NGS assays through variant review, QC, and troubleshooting in a high-complexity diagnostic setting.
- Contribute to workflow improvements and documentation to enhance reliability, reproducibility, and turnaround time.
- Collaborate with cross-functional teams to align assay needs with data and tooling requirements.

NGS · CLINICAL GENOMICS · VARIANT REVIEW

Clinical Bioinformatics Analyst III
Feb 2025 – Apr 2026

Foundation Medicine · Boston, MA

- Led advanced analysis and delivery of NGS data in a high-complexity clinical diagnostic laboratory as part of the Genomic Analysis and Reporting (GAR) team.
- Developed annotated genomic profiles and reviewed complex somatic variant calls for cancer patients across a broad range of tumor types.
- Implemented process improvements, including database search features and automation tools for bioinformatics workflows (Python, SQL, internal tooling).

CANCER GENOMICS · GAR · PYTHON · WORKFLOW AUTOMATION · AGENTIC CODING

Clinical Bioinformatics Analyst II / I
Sep 2017 – Feb 2025

Foundation Medicine · Cambridge, MA

- Developed annotated genomic profiles of somatic aberrations based on NGS data in cancer-patient samples.
- Reviewed genomic variant calls using IGV and proprietary tools in collaboration with scientific teams.
- Delivered clinical reports in compliance with SOPs and defined turnaround times.

SOMATIC VARIANTS · IGV · SOPS

Quality Control Analyst I
2016 – 2017

Sanofi Genzyme · Microbiology

- Performed QC testing in a regulated GMP environment, ensuring product safety and compliance with SOPs.
- Developed an analytical mindset for data integrity, documentation rigor, and process consistency that carried forward into bioinformatics work.

GMP · QC · REGULATED ENVIRONMENT

Research Assistant & NHGRI Internship
2013 – 2017

Colby College & National Human Genome Research Institute

- Early genomics and bioinformatics experience: genome sequence analysis, gene expression studies, and wet-lab molecular biology techniques.
- Contributed to research that led to two peer-reviewed publications (Blood 2019, JoVE 2017).

GENOMICS · RESEARCH · NHGRI

SKILLS

Programming

PYTHON · SQL · BASH · JAVA · R

Bioinformatics

CANCER GENOMICS · NGS · VARIANT ANALYSIS · IGV

Data & Workflows

DATA ANALYSIS · QC METRICS · WORKFLOW DESIGN

AI & Agentic Coding

CLAUDE CODE · AGENTIC WORKFLOWS · PROMPT DESIGN · AI CODE VALIDATION

Collaboration

MENTORING · TRAINING · PRESENTATIONS · DOCUMENTATION

Deepening software engineering skills: testing, workflow orchestration, and AI-assisted data tooling.

EDUCATION

B.A., Computational Biology

Minor: Mathematics

Colby College · Waterville, ME

2013 – 2017

- Capstone: Bioinformatic analysis of whole-genome sequence data from Diamond Blackfan Anemia patients to determine putative intronic mutations.

PUBLICATIONS

Whole-genome sequencing to detect small deletions in ribosomal protein genes in Diamond Blackfan Anemia patients.

Analyzed whole-genome data from DBA patients to determine putative intronic mutations. Blood (2019).

Measuring gene expression in bombarded barley aleurone layers with increased throughput.

Studied effects of TaABF1 serine phosphorylation on regulation of HVA1 and Amy32b. JoVE (2017).